

Freeze ablation – results

Three runs sharing the same setup (HCP-only, T=1200, temporal_kernel=20, base_lr=3e-4, effective batch 16, 20 epochs, DINOv2 ViT-S/14 reg4 ImageNet init). Only the parameter-freezing policy differs.

Architecture: what we added on top of DINOv2

We keep the DINOv2 SSL algorithm (student-teacher multi-crop self-distillation with iBOT masked-patch prediction) unchanged. Four fMRI-specific pieces are added on top of the official code:

1. **3D+1D patchify** (PatchEmbed3DPlus1D). Replaces DINOv2's 2D Conv2d patch embed. A hierarchical encoder built from Conv3Plus1d blocks (separable spatial-3D + temporal-1D conv, in the MovieGen-TAE style). Input (T, X, Y, Z) -> output token grid ($T_{\text{eff}} = T / \text{temporal_kernel}$, $N_{\text{spatial}} = N_x * N_y * N_z$). For HCP at T=1200 with temporal_kernel=20 and (45, 54, 45) voxels at patch_size=9: yields $60 * 150 = 9000$ patch tokens, plus 1 CLS + 4 register tokens = 9005 tokens fed to the ViT.
2. **Factorised positional embedding** (PositionEmbedding3D). Three small tables – pos_temporal (T_{eff}), pos_spatial (N_{spatial}), pos_cls (1) – broadcast and summed. $O(T_{\text{eff}} + N_{\text{spatial}})$ parameters instead of $O(T_{\text{eff}} * N_{\text{spatial}})$ for a flat 2D ViT pos table.
3. **3D multi-crop augmentation** (MultiCrop3D). For each scan, picks 2 global + 8 local random 3D spatial sub-volumes (scale [0.32, 1.0] and [0.05, 0.32] of the volume) and trilinearly resizes each back to (X, Y, Z). No photometric augmentation (color jitter, blur, solarize, ImageNet normalize) and no flip (preserves L/R brain asymmetry) – those don't apply to z-scored fMRI volumes.
4. **fMRI datasets** (HCPFullScanDataset, ADNIFullScanDataset, MixedFMRIDataset). Each scan loaded as (T, 1, X, Y, Z) with per-frame z-score. HCP scans are reconstructed from saved windowed .pt files; ADNI scans are slices of a shared 5D tensor.

Everything else (the SSL training loop, EMA teacher, DINO + iBOT + KoLeo losses, FSDP wrapping, sinkhorn-knopp centering, schedulers) is the official DINOv2 code, **unchanged**.

Strategies and trainable parameter counts

Backbone counts come from the actual training logs (backbone trainable=... line printed by apply_freeze_policy). The DINO head is a 3-layer MLP with 4096 prototypes and 256 bottleneck (~6.57 M params, computed analytically); it is random-init and **always trainable**. iBOT shares the DINO head (ibot.separate_head: false), no extra head.

Strategy	Backbone trained	Backbone trainable	Backbone frozen	DINO head	TOTAL trainable
A – Full fine-tune	all blocks + patch_embed + norm	36,212,256	0	6,566,144	42,778,400
B – Freeze last 3	patch_embed + blocks.9-11 + norm	16,776,480	19,435,776	6,566,144	23,342,624
C – Freeze except input	patch_embed only	11,450,016	24,762,240	6,566,144	18,016,160

The backbone contains the PatchEmbed3DPlus1D we added (~11.45 M – much bigger than a 2D ViT patch embed because of the temporal Conv3Plus1d ResBlocks), 12 transformer blocks (~21.2 M total), and a dead-weight 2D pos_embed allocated by the ViT parent class (~3.46 M, unused in fMRI mode but still allocated).

Best checkpoint per strategy (selected among available iters: 5999/8999 for A, 11999/17999 for B and C). Note on A: the original run (dinov2_fmri_20260524_144327) stopped at iter 10,410 instead of completing the 20 epochs (19,999 iters), so its results aren't strictly at the same training depth as B and C. A is being re-run to completion (fmri_vits_hcp_baseline.yaml).

HCP probes

Label	Metric	MLP baseline (step5)	A (full FT)	B (freeze last3)	C (freeze fmri)
Sex	AUC	0.935	0.751	0.696	0.844
BrainVol	MAE	-	116,179	117,744	99,455
GrayMatterVol	MAE	-	34,762	34,815	29,274
Age	MAE	4.228	3.044	3.056	2.931
FluidIntel	MAE	-	4.047	4.035	4.003
ProcSpeed	MAE	-	16.343	16.401	16.207
WorkingMem	MAE	-	10.442	10.410	10.411

(The HCP step5 MLP baseline run only covered Sex and Age; dashes mean it wasn't re-run for the remaining labels.)

ADNI probes

Label	Metric	MLP baseline (step5)	A (full FT)	B (freeze last3)	C (freeze fmri)
Sex	AUC	0.806	0.653	0.696	0.813
CDR	AUC	0.559	0.542	0.522	0.532
Degradation1Y	AUC	0.496	0.581	0.543	0.559
Degradation2Y	AUC	0.523	0.533	0.456	0.533
Degradation3Y	AUC	0.465	0.528	0.468	0.544
MMSE	MAE	4.492	2.478	2.426	2.750
Age	MAE	7.78	5.113	4.993	5.703

Bold = best value in the row. AUC: higher is better. MAE: lower is better. The MLP baseline column is the 3-layer MLP on raw flattened time-series (step5 in the FAIR repo, 5-fold CV; subject-aware StratifiedGroupKFold / GroupKFold for ADNI).

Source data: outputs/probes/probe_{adni,hcp}_<run_name>_iter<iter>.json – generated by slurm_jobs/probe_{adni,hcp}.sh, aggregated by python3 scripts/summarize_probes.py. Baselines from FAIR/logs/step5_{hcp,adni}.out.